Audrey Reinhard

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EDUCATION

B.S + M.S. in Computer Engineering **GPA - 3.72/4.00** University of California, Riverside June 2027
Honors: UC Regents Scholar, NSF S-STEM Scholarship
Relevant Coursework: Embedded Systems, Electronic Circuits, Logic Design, Computer Architecture, GPU Programming

SKILLS

Programming : JavaScript, Python, Java, C++, C, SQL,, PHP, Verilog,
Embedded Systems: Arduino, ESP32, Raspberry Pi, Actuators, Control Systems, UART, SPI, I2C, CAN, Sensors, Actuators
Robotics & AI: ROS2, OpenCV, TensorFlow, BERT, Sensor Fusion
Tools: Git, GitHub, Google Test, Oscilloscope, Logic Analyzer, Fusion 360
Web & APIs: REST APIs, MySQL, HTML, CSS

INTERN EXPERIENCE

Full Stack Development Intern The Three Logistics, Riverside, CA February 2024 – May 2024

- Developed backend features using **REST APIs**, enabling address verification, suggestions, and autocomplete functionality
- Improved shipment processing speed by 25% through optimized **API- based** address validation workflows
- Implemented package tracking and account linking using a **MySQL database**, supporting secure data storage and retrieval
- Built responsive user interfaces and dashboards using **JavaScript, PHP, HTML, and CSS**

ENGINEERING EXPERIENCE

Robotics Lab Student Lead Robotics Lab**,** UC Riverside September 2024 – Present

- Led development of embedded robotics systems integrating **Arduino, Raspberry Pi, sensors, and motor controllers**, supporting multiple lab research and instructional projects.
- Designed and implemented a magnetic-stripe access and equipment tracking system integrating **SQL databases**, G**oogle Sheets API**, and hardware interfaces; significantly improved lab resource tracking and utilization.
- Designed and taught a 6-week Robotics Camp covering **Python control loops**, **OpenCV**, **serial communications**, **Convolutional Neural Networks**, and embedded systems fundamentals.

Data Science Fellow *DZMaC Lab,* National Science Foundation, UC Riverside June 2024 – July 2024

- Fine-tuned a **DistilBERT LLM classifier** to categorize neuroscience transcripts with 83% accuracy.
- Automated audio preprocessing (noise reduction, segmentation), cutting transcription time by 72%.
- Collaborated with neuroscience researchers to integrate model outputs into cognitive recall studies.

PROJECTS

Autonomous Rover Navigation System | Github **ESP32-S3 + NavQPlus + ATmega328P |** Oct 2025 – February 2026

- Implemented a **ROS2 multi-node robotic navigation system** coordinating sensing, perception, and motion control.
- Developed an **ESP32-based** sensor hub integrating **ultrasonic**, **GPS**, **IMU**, and **IR** sensors, transmitting real-time data to NavQPlus via **CAN bus.**
- Implemented **GIS path planning** and **sensor fusion** to generate navigation routes across mapped campus environments.
- Integrated **NPU-accelerated image segmentation** to improve obstacle awareness and environmental perception.
- Programmed **bare-metal C motor drivers** on Arduino and designed the full robot chassis in **Fusion 360.**

Oura Ring Tamagotchi, Github **Raspberry Pi Pico + FPGA |** May 2026 – June 2026

- Engineered a wearable-integrated embedded system that synchronizes **Oura Ring API** sleep, activity, and readiness data with a Tamagotchi-style game running on a Raspberry Pi Pico.
- Designed a high-speed **SPI** interface between a Raspberry Pi Pico and FPGA driving a **480×270 TFT display**, with graphics rendered using **LVGL**.
- Developed FPGA display logic and embedded firmware in **Verilog** and **C** to support real-time animations and health-data visualization.

Secret Journal, Citrus Hacks Github **Python** + **ATmega328P** + **DeepFace |** April 18-19, 2026

- Developed a secure smart journal that authenticates users using **facial and voice recognition**, building a Python backend with **DeepFace** and **Resemblyzer** integrated into a web interface using **HTML, CSS, and JavaScript**.
- Programmed an **Arduino Nano** to control a **stepper motor locking mechanism**, status LEDs, and a **16×2 LCD**, establishing **UART** communication between the embedded hardware and the Python backend.
- Designed the journal enclosure in **Fusion 360** and integrated **Gemma (Gemini)** to generate personalized writing prompts based on authenticated user profiles.

Computer Vision Motion Detection , Github **Python** + **YOLO** + **OpenCV |** February 2026 – March 2026

- Developed a real-time computer vision system to detect, classify, and track **pedestrians, cyclists, and scooters** from live video streams using a custom-trained **YOLO** model.
- Built a **TensorFlow** training pipeline with data augmentation, validation, and custom dataset fine-tuning to improve object detection accuracy across multiple mobility classes.
- Combined **classical computer vision** techniques—including frame differencing, grayscale thresholding, morphological filtering, contour extraction, and bounding-box overlap analysis—with **YOLO** detections to identify moving objects and reduce false positives in real time.

Space Invaders AVR Game **|** Github **ATmega328P** | May 2025 – June 2025

- Built a real-time game in **embedded C** on an ATmega1284 microcontroller using a **synchronous task schedule**
- Programmed **SPI**-driven ST7735 LCD, joystick **ADC** sampling, and **PWM**-driven audio.
- Reduced LCD tearing by optimizing **SPI** bandwidth and implementing partial screen redraw logic.

Chronotype-Based Task Scheduler, Github February 2025 – March 2025

- Developed a **C++/Qt** application that schedules user tasks based on chronotype quiz results and availability.
- Implemented modular **SOLID**-compliant design with **GoogleTest** unit testing
- Collaborated in a 4-member **Agile** team using **GitHub** for **sprint** management.